PigeonTrack™

Enterprise Carrier Pigeon Fleet Management System

System Administration & Operations Manual

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1. Introduction

PigeonTrack™ is an enterprise-grade fleet management system for organizations that operate carrier pigeon networks as a secondary communication channel alongside conventional IT infrastructure. While this may raise eyebrows in the average board meeting, the use cases are entirely legitimate: regulatory compliance in environments with strict out-of-band communication requirements, air-gapped segment bridging, disaster recovery signaling when all fiber has been severed, or simply fulfilling contractual obligations with clients who are, diplomatically speaking, cautious about digital modernity.

PigeonTrack manages the full operational lifecycle of a pigeon fleet: GPS leg-band telemetry, flight trajectory optimization, feed inventory (including SAP-integrated millet procurement), veterinary scheduling, and cryptographically signed message capsule tracking. The system exposes a modern REST API and a React-based dashboard for dispatchers, supervisors, and pigeon welfare officers.

This manual covers system architecture, operational requirements, and — critically — the network security prerequisites that your IT Security team will need before the system can be deployed.

1.1 Target Audience

This document is intended for:

- System administrators responsible for server deployment and maintenance
- Network security engineers configuring perimeter firewall rules
- Operations staff using the web-based dispatch interface
- Pigeon welfare officers using the health monitoring module
- SAP Basis consultants who will, regrettably, also be involved

> ⚠ **Important Note**
>
> This manual assumes the reader understands basic network concepts such as TCP/IP, DNS, and firewall zones.
>
> It does not assume familiarity with carrier pigeons. The necessary avian background is covered in Appendix C.

1.2 Document History

Version	Date	Author	Change Summary
3.7.2	2025-06-01	A. Taube	Firewall section expanded; SSH key rotation procedure added
3.7.0	2025-04-15	A. Taube	SAP RFC connector documented; millet procurement workflow

3.5.1	2025-01-10	F. Brieftaube	DMZ architecture revised; auto-update mechanism documented
3.0.0	2024-09-01	F. Brieftaube	Initial release of v3 architecture

2. System Overview

PigeonTrack operates as a web application hosted in the organization's DMZ (Demilitarized Zone). This placement is deliberate: the system must be reachable both from the internal corporate network and from the public internet — dispatchers in the field need to assign missions to pigeons without being physically present on-premises, and external integration partners (primarily certified lofts and veterinary networks) require API access.

2.1 Architecture

The system consists of the following components:

Component	Role	Location
PigeonTrack Web Application	Primary user interface and API gateway	DMZ
PostgreSQL Database Cluster	Persistent storage for fleet, mission, and telemetry data	Internal Server Zone
SAP ERP Integration	Procurement of feed materials; financial cost center allocation	Internal SAP Zone
Update Service	Automated retrieval of firmware and application updates	DMZ → Internet
SSH Management Access	Remote administration by IT Operations staff	Internal → DMZ

The architecture follows a strict separation-of-concerns model. The DMZ server holds no persistent business data itself — all state is stored in the internal PostgreSQL cluster. This means that a compromise of the DMZ host does not directly expose the data tier, provided the database firewall rules are correctly configured.

2.2 Network Zone Model

PigeonTrack operates across three network zones:

- INTERNET — Public networks; untrusted. External users, field dispatchers, and integration partners access the web UI from here.
- DMZ — Semi-trusted perimeter zone. The PigeonTrack application server resides here. It has no direct internet access for inbound connections but may initiate outbound connections for update retrieval.
- INTERNAL — Trusted corporate network. Contains the PostgreSQL database servers, the SAP application layer, and the workstations of IT Operations staff who administer the DMZ host.

Traffic flows are strictly directional. No traffic is permitted to flow directly from INTERNET to INTERNAL or vice versa. All such traffic is mediated through the DMZ application layer.

3. Functional Description

3.1 Core Modules

PigeonTrack is organized into six functional modules:

Module	Description
Fleet Registry	Master database of all registered pigeons, including breed, age, flight endurance, and special dietary requirements.
Mission Dispatch	Interface for creating, assigning, and tracking message delivery missions. Supports priority levels: ROUTINE, URGENT, and EXTREMELY URGENT (requires two-person authorization).
Telemetry Dashboard	Real-time visualization of GPS leg-band data. Includes wind-adjusted ETA calculations and automated deviation alerts.
Health & Welfare	Veterinary appointment scheduling, inoculation tracking, and psychological status assessment (yes, pigeons experience stress).
Feed & Supply Chain	Inventory management for feed materials with SAP integration for automated purchase order generation.
Reporting & Audit	Mission completion reports, fleet utilization statistics, and regulatory compliance exports.

3.2 User Roles

- Dispatcher — Creates and monitors missions; read access to fleet and telemetry data.
- Fleet Manager — Full access to fleet registry and health module; approves URGENT missions.
- Supply Officer — Manages feed inventory and SAP procurement workflows.
- System Administrator — Full administrative access; no access to mission content (separation of duties).
- Pigeon Welfare Officer — Read access to health data; can raise welfare flags that block mission dispatch.

3.3 SAP Integration

PigeonTrack integrates with SAP ERP via RFC (Remote Function Call) for two purposes:

1. Feed procurement: When millet, sunflower seed, or grit inventory falls below configured thresholds, PigeonTrack generates a purchase requisition in SAP automatically. This prevents the operationally catastrophic scenario of a pigeon fleet grounded due to insufficient nutrition.
2. Cost center allocation: Mission costs (depreciation, feed, veterinary) are posted to the requesting department's SAP cost center for internal chargeback.

The RFC connection uses the SAP standard port 3300 (for instance 00). The PigeonTrack service account in SAP requires authorization for function modules MM_PURCHASE_REQUISITION_MAINTAIN and CO_DOCUMENT_POST only. No broader SAP authorizations are granted.

4. Deployment

4.1 Server Requirements

Parameter	Minimum	Recommended
CPU	4 vCPU	8 vCPU
RAM	8 GB	16 GB
Disk (OS + App)	40 GB SSD	80 GB SSD
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
Runtime	Docker 24+	Docker 26+ with Compose V2
Network Interfaces	1 (DMZ)	2 (DMZ + dedicated mgmt)

4.2 Installation

PigeonTrack ships as a Docker Compose stack. Installation on a freshly provisioned DMZ host:

```
# Clone the deployment repository
git clone https://deploy.pigeontrack.io/pigeontrack.git
cd pigeontrack
# Configure environment variables
cp .env.example .env && nano .env
# Launch the stack
docker compose up -d
```

4.3 Automatic Updates

PigeonTrack includes an integrated update agent that checks for new releases every 24 hours. The agent contacts the PigeonTrack update server at updates.pigeontrack.io over HTTPS (TCP/443). It downloads, verifies (SHA-256 checksum + GPG signature), and applies updates automatically during the configured maintenance window (default: Sunday 02:00–04:00 local time).

> ⚠ **Security Requirement**
>
> The update agent initiates outbound connections only. No inbound connections from the update server are required.
>
> The destination hostname is updates.pigeontrack.io (resolves to static IPs documented in Section 5.3).
>
> If HTTPS inspection (TLS intercept) is active on your perimeter, the update agent's certificate pinning will fail.

Either exempt the update traffic from inspection, or disable automatic updates and apply them manually.

5. Network Security Requirements

This section defines all firewall rules required for PigeonTrack to function correctly. These rules must be implemented before the system goes live. The rules are grouped by traffic flow direction and functional purpose.

This section is the primary input for a firewall change request. Each rule is described with sufficient detail for a network security engineer to implement it without ambiguity.

5.1 Architecture Assumptions

- The PigeonTrack application server has the DMZ IP address 10.20.0.50 (assign to actual IP before submitting change request).
- Internal PostgreSQL servers are reachable at 10.10.5.10 and 10.10.5.11 (primary and read replica).
- The SAP application server is at 10.10.10.20 (SAP System ID: PT1, Instance 00).
- Internal user workstations occupy the subnet 10.10.0.0/16.
- The firewall's external interface accepts traffic from 0.0.0.0/0 for internet-facing rules.

5.2 Firewall Rules — Summary Table

The following table provides a complete summary of all required rules. Detailed justification for each rule follows in Section 5.3.

Rule ID	Name	Source Zone	Destination Zone	Source	Destination	Port / Protocol	Action
FW-PT-01	Internal Web Access	INTERNAL	DMZ	10.10.0.0/16	10.20.0.50	TCP/443, TCP/80	ALLOW
FW-PT-02	Internet Web Access	INTERNET	DMZ	0.0.0.0/0	10.20.0.50	TCP/443	ALLOW
FW-PT-03	App → PostgreSQL (Primary)	DMZ	INTERNAL	10.20.0.50	10.10.5.10	TCP/5432	ALLOW
FW-PT-04	App → PostgreSQL (Replica)	DMZ	INTERNAL	10.20.0.50	10.10.5.11	TCP/5432	ALLOW
FW-PT-05	App → SAP RFC	DMZ	INTERNAL	10.20.0.50	10.10.10.20	TCP/3300	ALLOW
FW-PT-06	Auto-Update Outbound	DMZ	INTERNET	10.20.0.50	updates.pigeontrack.io	TCP/443	ALLOW

FW-PT-07	SSH Management	INTERNAL	DMZ	10.10.1.0/24	10.20.0.50	TCP/22	ALLOW

TCP/80 in rule FW-PT-01 is used for HTTP-to-HTTPS redirect only. The application does not serve content over plain HTTP. If your security policy prohibits any TCP/80 rule, the redirect can be disabled and users must connect directly via HTTPS.

5.3 Rule Justifications

FW-PT-01 — Internal User Web Access

Business units and internal operational staff access the PigeonTrack web interface via browser. This includes dispatchers assigning missions from their desks, fleet managers reviewing telemetry, and supply officers approving SAP purchase requisitions. HTTPS (TCP/443) is the primary protocol. TCP/80 is accepted for redirect to HTTPS only.

- Source: Corporate user network (10.10.0.0/16)
- Destination: PigeonTrack DMZ server (10.20.0.50)
- Ports: TCP/443 (primary), TCP/80 (redirect only)
- Direction: INTERNAL → DMZ

FW-PT-02 — Internet Web Access

Field dispatchers, partner loft operators, and the PigeonTrack mobile application connect from the public internet. All external access is HTTPS-only. HTTP access from the internet is not permitted. If a reverse proxy or WAF is deployed in front of the DMZ server, this rule targets the proxy's DMZ IP instead.

- Source: Any (0.0.0.0/0)
- Destination: PigeonTrack DMZ server (10.20.0.50)
- Ports: TCP/443 only
- Direction: INTERNET → DMZ

Security Recommendation

Consider restricting source IPs to known field dispatcher ranges if operational context allows.

A WAF in front of this rule is strongly recommended for production deployments.

Rate limiting at the application layer is implemented but should be supplemented at the network level.

FW-PT-03 & FW-PT-04 — Application to PostgreSQL

PigeonTrack stores all mission data, fleet registry, telemetry history, and user accounts in a PostgreSQL cluster. Two separate rules are required: one for the primary database server and one for the read replica, which serves reporting queries and the telemetry dashboard to reduce load on the primary.

- Source: PigeonTrack application server (10.20.0.50)

- Destinations: 10.10.5.10 (primary), 10.10.5.11 (read replica)
- Port: TCP/5432 (PostgreSQL default)
- Direction: DMZ → INTERNAL

These rules intentionally restrict the source to the single application server IP. No subnet-level access to the database is granted.

FW-PT-05 — Application to SAP RFC

The feed procurement and cost center allocation integration uses SAP RFC over TCP/3300. This is the standard RFC port for SAP instance 00. If your SAP instance uses a different instance number (01, 02, etc.), the port is 3300 + instance number (e.g., instance 01 uses TCP/3301).

- Source: PigeonTrack application server (10.20.0.50)
- Destination: SAP application server (10.10.10.20)
- Port: TCP/3300 (SAP RFC, instance 00)
- Direction: DMZ → INTERNAL

> ⚠ **Verify Before Implementing**
>
> Confirm the SAP instance number with your SAP Basis team before submitting the firewall change request.
>
> Granting access to the wrong port, or to a broader SAP port range, constitutes a significant security risk.

FW-PT-06 — Automatic Update Outbound

The PigeonTrack update agent connects outbound to updates.pigeontrack.io on TCP/443 to retrieve and verify software updates. This is a client-initiated, outbound-only connection. The update server never initiates inbound connections.

- Source: PigeonTrack application server (10.20.0.50)
- Destination: updates.pigeontrack.io (FQDN-based rule recommended; static IPs: 185.100.4.20, 185.100.4.21)
- Port: TCP/443 (HTTPS with certificate pinning)
- Direction: DMZ → INTERNET

If your firewall platform supports FQDN-based destination rules (e.g., Palo Alto FQDN objects), prefer that over static IPs to accommodate future IP changes by the update infrastructure. If static IPs are required, monitor PigeonTrack release notes for infrastructure change announcements.

FW-PT-07 — SSH Management Access

System administrators manage the PigeonTrack server via SSH. Access is restricted to the IT Operations management subnet (10.10.1.0/24). Under no circumstances is SSH access to the DMZ host permitted from the internet.

- Source: IT Operations subnet (10.10.1.0/24) — not the entire internal network
- Destination: PigeonTrack application server (10.20.0.50)
- Port: TCP/22 (SSH)
- Direction: INTERNAL → DMZ

Security Best Practice

The source range 10.10.1.0/24 should be restricted to the IT Operations jump host if one exists.

SSH key-based authentication is mandatory. Password authentication is disabled in the PigeonTrack server configuration.

SSH access logs are forwarded to the central SIEM via syslog. See Section 6 for log forwarding configuration.

5.4 Rules Explicitly NOT Required

For completeness, the following traffic flows are explicitly not required and must not be opened:

- Inbound SSH from INTERNET to DMZ — Never required. Field administrators use VPN + SSH.
- Direct INTERNET → INTERNAL access — No scenario in PigeonTrack requires bypassing the DMZ.
- DMZ → INTERNAL on any port other than TCP/5432 and TCP/3300 — The application has no other internal dependencies.
- Inbound connections from update server — Update retrieval is outbound-only.
- ICMP (ping) — Not required for application function. Allow or deny per your standard policy.

6. Operations & Administration

6.1 Log Forwarding

PigeonTrack generates structured JSON logs for all application events, authentication attempts, and API calls. Logs are written to stdout and captured by Docker's logging driver. Forward them to your SIEM using the syslog driver or a log shipping agent.

Critical log categories:

- auth — Login attempts, session creation, privilege escalation
- mission — Mission creation, dispatch, completion, cancellation
- api — All inbound API requests with response codes
- update — Update check results, download events, applied versions
- db — Connection pool status, query errors (no query content is logged)

6.2 SSH Key Management

SSH access to the DMZ host uses ED25519 keys only. RSA keys below 4096 bits are rejected by the SSH daemon configuration. Each administrator uses a personal key pair. Shared keys are prohibited.

Key rotation procedure:

3. Generate a new key pair on the administrator workstation: ssh-keygen -t ed25519 -C 'admin@pigeontrack'
4. Submit the public key to the IT Operations ticket system for review.
5. An approving administrator adds the new key to ~/.ssh/authorized_keys on the DMZ host.
6. The old key is revoked and removed within 24 hours of the new key being confirmed functional.

6.3 Health Checks

The application exposes a health endpoint at /health (HTTP/200 when healthy). This endpoint is unauthenticated and intended for load balancer probes and monitoring systems. It returns no sensitive information. The endpoint is not subject to rate limiting.

7. Known Limitations & Caveats

A Note on Realistic Expectations

PigeonTrack is a serious enterprise system. The following limitations are documented honestly, in the spirit of RFC 1925: 'In protocol design, perfection has been reached not when there is nothing left to add, but when there is nothing left to take away — except the pigeons.'

- Latency — Carrier pigeon message delivery latency ranges from 15 minutes to 3 days depending on weather, predators, and pigeon motivation. PigeonTrack's real-time dashboard accurately reflects this reality. SLA definitions should be adjusted accordingly.
- Throughput — The theoretical maximum message throughput of a pigeon fleet is bounded by biology. PigeonTrack does not solve this problem. It makes it more visible, which is arguably worse.
- SAP Integration Edge Case — If a purchase requisition is submitted for 'emergency grit' during an SAP month-end close, it will be queued and processed after the close. This is a SAP limitation. PigeonTrack raises an alert. The pigeons will have to wait.
- GPS in Urban Canyons — Leg-band GPS accuracy degrades in dense urban environments. Telemetry tracks in city centers should be interpreted with a margin of ±200 meters. This is also, not coincidentally, a documented limitation of the pigeons themselves.
- Auto-Update During High Traffic — The update maintenance window should be configured for periods of low operational activity. Applying an update while 40 missions are in flight is technically safe (zero-downtime deployment) but operationally inadvisable. The pigeons will land regardless.

Appendix A: Firewall Change Request Template

The following template may be submitted directly to the network security team as a change request. Fill in the bracketed fields with environment-specific values before submission.

Field	Value
Request Title	PigeonTrack v3.7 — New Application Firewall Rules
Requestor	[Name, Department]
Application Owner	[Fleet Manager Name]
Target Go-Live Date	[Date]
Change Risk	Standard
Rules Requested	FW-PT-01 through FW-PT-07 as defined in Section 5.2
Rollback Plan	Remove all 7 rules. Application will be inaccessible but no data is at risk.
Test Procedure	Access https://[DMZ IP] from internal browser; verify login page loads.
Security Contact	[IT Security contact for queries]

Appendix B: Port Reference

Port	Protocol	Service	Direction	Notes
TCP/80	HTTP	Web (redirect)	Inbound to DMZ	Redirect to HTTPS only; optional
TCP/443	HTTPS	Web UI & API	Inbound to DMZ	Primary application port
TCP/443	HTTPS	Auto-Update	Outbound from DMZ	To updates.pigeontrack.io only
TCP/5432	PostgreSQL	Database	DMZ → Internal	Standard PostgreSQL port
TCP/3300	SAP RFC	ERP Integration	DMZ → Internal	Instance 00; adjust for other instances
TCP/22	SSH	Administration	Internal → DMZ	IT Operations subnet only

Appendix C: Essential Pigeon Background

For readers unfamiliar with carrier pigeon operations:

- Carrier pigeons (Columba livia domestica) navigate using a combination of magnetic field sensing, solar positioning, and familiarity with landmarks. They do not use GPS themselves — that is what PigeonTrack's leg bands are for.
- A healthy racing pigeon can cover 500–800 km in a single flight at speeds of 80–130 km/h. Performance degrades in headwinds, heavy rain, and Mondays.
- Pigeons are homing animals: they fly toward their home loft, not toward an arbitrary destination. The operational model is therefore: train pigeon at source, deploy pigeon at destination, release pigeon to return to source carrying the message. This is not a design flaw. It is simply how pigeons work.
- Millet is preferred. Corn is acceptable. Sunflower seeds are a luxury item and should not appear in routine purchase requisitions without Fleet Manager approval.

For further reading, the authors recommend RFC 1149 (A Standard for the Transmission of IP Datagrams on Avian Carriers) and its quality-of-service extension RFC 2549. Both are genuine IETF documents. This is not a drill.

PigeonTrack™ is a trademark of PigeonTrack GmbH & Co. KG. All pigeon welfare standards are in compliance with the EU Animal Transport Regulation (EC) No 1/2005, as interpreted by our legal team's most creative associate.